Empirical Resource Profiling & Asymptotic Computational Complexity Benchmark

Model Architecture	Parameters	Disk Storage	Train Time	Latency	Test Acc	Space Complexity
Logistic Regression (SGD)	32	1.12 KB	1 ms	0.06 ms	93.94%	$\mathcal{O}(D)$
SVM (RBF Kernel)	1,921	16.87 KB	3 ms	0.06 ms	90.91%	$\mathcal{O}(N_{sv} \cdot D)$
Random Forest (150 Trees)	3,328	302.24 KB	135 ms	12.90 ms	90.91%	$\mathcal{O}(T \cdot \text{Nodes})$
XGBoost (60 Rounds)	682	71.94 KB	91 ms	0.16 ms	93.94%	$\mathcal{O}(T \cdot \text{Depth})$
Neural Net (MLP 64-32)	4,161	135.46 KB	17 ms	0.09 ms	93.94%	$\mathcal{O}(H_1 \cdot H_2)$
Method 1: Angle VQC (PCA 6-D)	**38**	**0.45 KB**	**48.20 s**	**6.06 ms**	**81.82%**	$\mathcal{O}(n)$ Qubits
Method 2: Amplitude VQC (64-D)	**38**	**0.45 KB**	**52.40 s**	**6.04 ms**	**84.85%**	$\mathcal{O}(\log_2 D)$ Qubits